

The Long Island Green Homes Program

It all started early last December with a phone call from a homeowner who wanted an estimate on insulating his home. My son Josh and I are building performance contractors and do Energy Star home performance work on existing residential buildings. But this wasn't the typical call. The voice on the other end was Steve Bellone, the Babylon town supervisor, who has been aggressively spearheading a green energy policy here on Long Island, New York, for the past several years—before it was in vogue. His town was the first to pass laws requiring that all new construction of residential buildings be Energy Star, and recently (in 2006) requiring that all new construction of commercial buildings over 4,000 square feet be LEED certified. So naturally I paid a little closer attention to what he was looking to do. He had gotten my name from another leading energy guru here on Long Island, Frazer Dougherty of North Fork Retrofit/Solar Options, who had given him a list of local performance contractors.

Walking the Walk

When I arrived for the initial visit he was there, not only with pen in hand and armed with a multitude of questions, but with Dorian Dale, the energy director for the town of Babylon—who, by the way, is the only energy director for all the municipalities on Long Island. At first I was a little intimidated by the attention, but I began to relax when they starting firing off questions about what I was proposing and how it would affect Bellone's energy costs. The questions were good ones. They were both very inquisitive about how I was going to determine the energy savings in relation to the cost. I didn't find out until later why they were asking me such specific

questions. I think they already knew then, although they didn't tell me until later, what they were up to.

It was around the end of January, when Josh and I had finished all the work on the Bellone home, that they approached us and asked us to help them in a program they were developing called LI Green Homes. The LI Green Homes program is the mother lode of energy conservation. Its focus is on retrofitting existing homes, which account for at least 40% of the country's energy use. Before he could talk the talk, Bellone wanted to walk the walk. I guess Bellone and Dale



Babylon Town Supervisor Steve Bellone is greening his city.

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first wanted to see if I performed the work according to what I had proposed.

Bellone's house was under renovation at the time I did the insulation work, and the second floor had been gutted. One of the concerns Frazer Dougherty had raised was that it was important to seal and insulate the band joist; and what made it more important was the fact that the second floor had been added to the house, and there was a 14-inch gap between floors. What we came up with was to remove the flooring on the second floor to access the band joist. This gave us the ability to seal the entire band joist with 6 inches of high-density foam.

We also sealed the rim joist in the basement with 6 inches of high-density foam, and installed 2 inches of Thermax on the basement walls to bring the basement into the thermal boundary with R-12 insulation. We then netted the second-floor walls and dense-packed with cellulose, giving

the walls an R-value of about 19. After that we air sealed and insulated the attic with loose-fill cellulose to around R-55. The house had another addition in the rear; we sealed and dense-packed the first-floor walls and lower attic of this addition to around R-30.

The initial blower door reading was around 5,400 CFM₅₀. After the work was completed, it was less than 2,400 CFM₅₀. If the first floor had been gutted, we could have reduced it below 2,000 CFM₅₀. Bellone had two 60,000 Btu gas-fired boilers and two 40-gallon gas water heaters in the basement, because the second floor had been an apartment. The calculations showed that we could remove one boiler and one water heater. I would calculate the energy savings to be around 40%–50%.

Steve's wife was not happy when he told her about the work we were going to do. She told him that it better be good, because they were having their first child at the end of January and the work had better be done on time, or else! So we expedited the project. The baby was born soon after the renovation was finished. Since then, his wife and daughter have been happy, and his wife can't believe how comfortable it is in the house.

Long Island Green Homes

We have been working with Dale for countless hours on the LI Green Homes program, and after 12 months it is almost ready for launch. It was a very challenging project. The program is unique to Long Island, and if it is

successful it may exponentially change the energy outlook not only here on Long Island, but perhaps throughout the nation. What's interesting is that Supervisor Bellone was really involved in forming the program.

His knowledge of the

subject was extensive, because he has been studying the issues for some time. This is uncommon, since most politicians know very little about the subject; usually they listen to what other



Dorian Dale, the Energy Director for Babylon, is changing the world one lightbulb at a time.

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jurisdictions are developing, and then follow their lead.

Here is how the program is going to operate. First, the law covering municipal waste fees charged to residents and businesses has been changed to include carbon waste based on its carbon content, such as the carbon emitted from heating systems into the atmosphere. This is an innovative idea, to say the least, and forms the basis for funding the program. A reserve fund in the waste program is going to finance energy retrofits in Babylon's housing stock, thereby reducing carbon waste and in the same stroke lowering energy costs for homeowners.

As a certified Building Performance Institute (BPI) contractor, I will go into the house and perform a detailed assessment, as well as the required safety testing. (The town of Babylon will require that the auditors and contractors be licensed, insured, and certified by BPI to ensure quality control.) I will then develop a work scope from the findings. This is where the difficult part comes in. The homeowner will be obligated to submit one to two years of energy bills so that we can estimate the savings from the retrofit work. This information will be put into a software program, not established as of yet, to pump out some numbers and determine the energy savings, the monetary savings, and the payback period. The measures that will be performed include air sealing, installing insulation, caulking, and replacing heating and domestic hot water systems. The program was not designed to finance whole-house renovations. We are just looking for the most cost-effective measures at the greatest value with the shortest payback. When the measures have been paid off, as I explain below, homeowners should be able to save enough money to pay for any further energy-saving measures on their own.

The cost of the work will range from \$4,000 to \$12,000, per house in the pilot program, as it seeks to extend to an optimal number of houses based on the money available. Additional funds may come from the state or the federal government if the program

Babylon was the first to pass laws requiring that all new construction of residential buildings be Energy Star.

is successful. The hope is that these measures will lower the typical home's energy costs by 25%–40% while reducing carbon waste. Dorian Dale says that the average home generates carbon waste equivalent to the carbon contained in two 13-lb bags of charcoal. Think what the energy savings will do for the environment and how many barrels of oil we can save!

Nor will homeowners incur any debt or obtain any line of credit; nor will their taxes be raised, since the obligation will be assigned to the house as a benefit assessment. The obligation will be structured so that the projected savings cover the capital costs. The debt will be arranged around the improvement package. For instance, if the energy savings calculations show savings of

The unique part of the LI Green Homes program is that homeowners will not pay for any of the measures up front, since the contractor will be paid from the solid waste fund.



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Josh Manning installs dense pack cellulose to exterior walls using the bib system.



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The dense pack cellulose in the completed wall should be around 3 1/2 lb per square foot.

green building



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Thermax is installed on the foundation wall. The rim joist was sprayed with 4 inches of high density foam to seal and insulate the thermal boundary between the first floor and the basement.

\$125 a month over an eight-year period, the homeowner will pay a benefit assessment of around \$100 a month. This will give the homeowner a \$25 positive cash flow each month; and after the assessment is finished, the homeowner will receive the full savings. If

the homeowner chooses to move before the benefit assessment is paid off, the cost will stay with the property and will be paid down by the new owner. This sounds to me like a win-win situation.

The town has already signed up over 250 residents who want to hear

more about the program. Dorian Dale will oversee the implementation of the program and provide quality assurance. With just a few minor details to work out, we expect to move forward with the LI Green Homes program shortly before this article goes to press. We should know by next summer how successful the program is. If it is successful—and I'm sure it will be—the program can be extended to other townships here on Long Island, and possibly to other areas of the country. **He**

—Rich Manning

Rich and Josh Manning have been in business with Energy Master & Environmental Solutions since 2004. Both are BPI certified contractors, and are accredited Zeradraft contractors. They are also certified to perform HERS ratings on new Energy Star homes, and their company is LEED AP certified.

For more information:

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